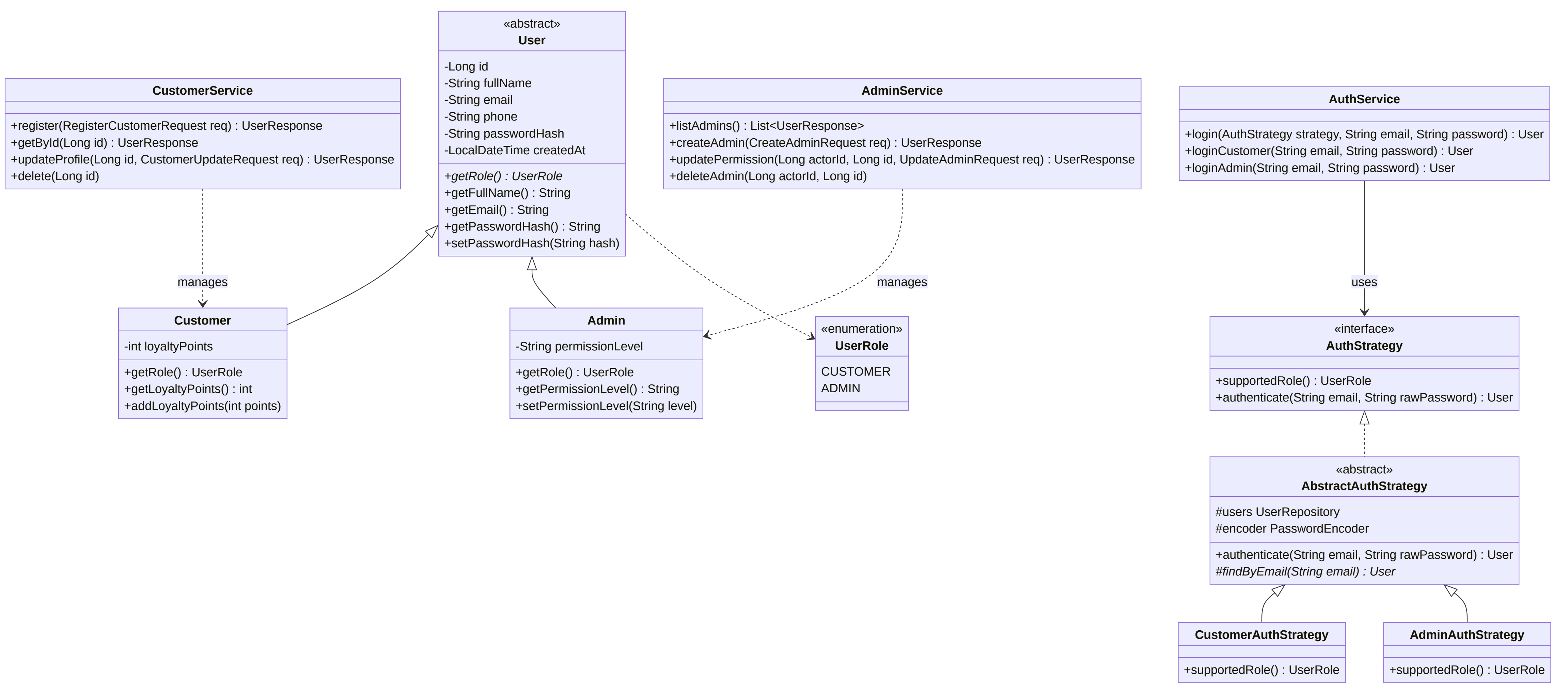
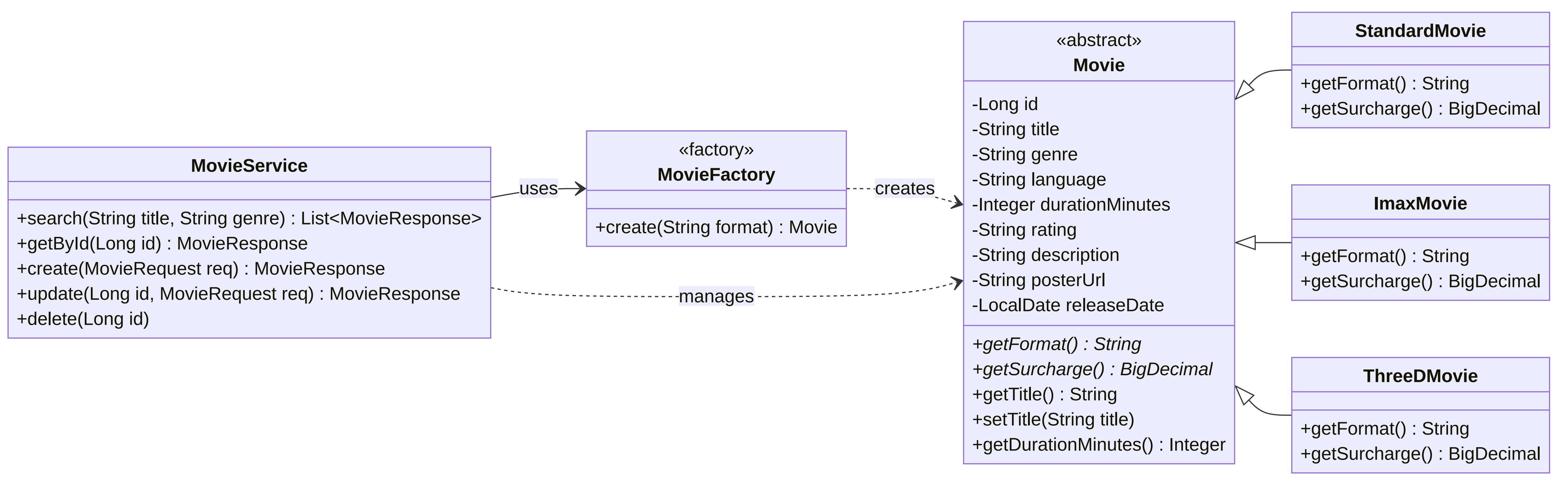
CineMax Movie Ticket Booking – Class Diagrams

SE1020 – Object Oriented Programming. Generated from docs/CLASS_DIAGRAM.md.

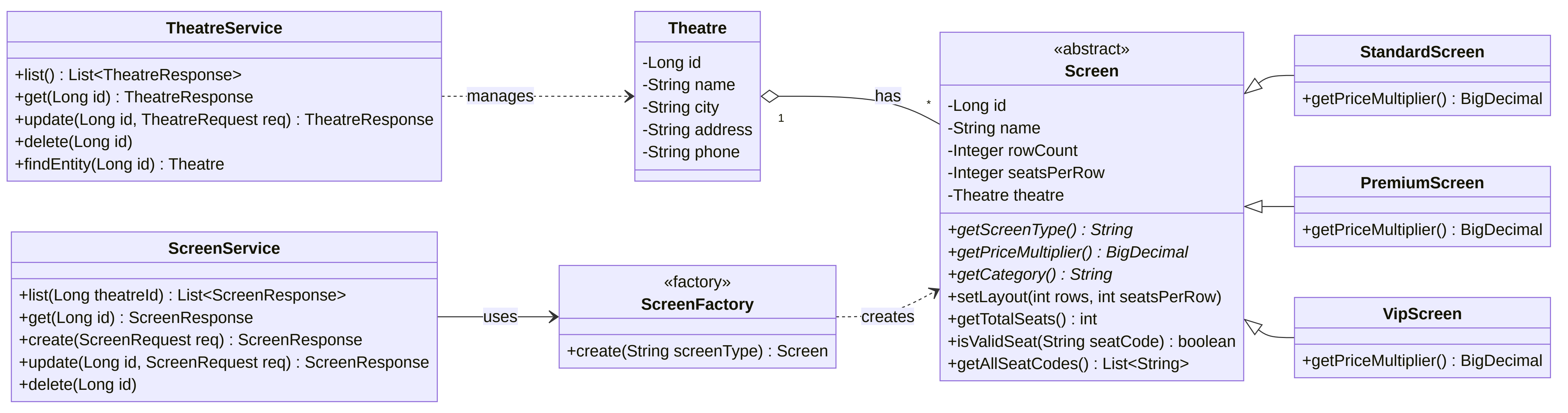
1. User & authentication (Member 1)


CustomerService
+register(RegisterCustomerRequest req) : UserResponse
+getById(Long id) : UserResponse
+updateProfile(Long id, CustomerUpdateRequest req) : UserResponse
+delete(Long id)
«abstract»
User
-Long id
-String fullName
-String email
-String phone
-String passwordHash
-LocalDateTime createdAt
+getRole() : UserRole
+getFullName() : String
+getEmail() : String
+getPasswordHash() : String
+setPasswordHash(String hash)
AdminService
+listAdmins() : List<UserResponse>
+createAdmin(CreateAdminRequest req) : UserResponse
+updatePermission(Long actorId, Long id, UpdateAdminRequest req) : UserResponse
+deleteAdmin(Long actorId, Long id)
AuthService
+login(AuthStrategy strategy, String email, String password) : User
+loginCustomer(String email, String password) : User
+loginAdmin(String email, String password) : User
manages
Customer
-int loyaltyPoints
+getRole() : UserRole
+getLoyaltyPoints() : int
+addLoyaltyPoints(int points)
Admin
-String permissionLevel
+getRole() : UserRole
+getPermissionLevel() : String
+setPermissionLevel(String level)
manages
«enumeration»
UserRole
CUSTOMER
ADMIN
uses
«interface»
AuthStrategy
+supportedRole() : UserRole
+authenticate(String email, String rawPassword) : User
«abstract»
AbstractAuthStrategy
#users UserRepository
#encoder PasswordEncoder
+authenticate(String email, String rawPassword) : User
#findByEmail(String email) : User
CustomerAuthStrategy
+supportedRole() : UserRole
AdminAuthStrategy
+supportedRole() : UserRole

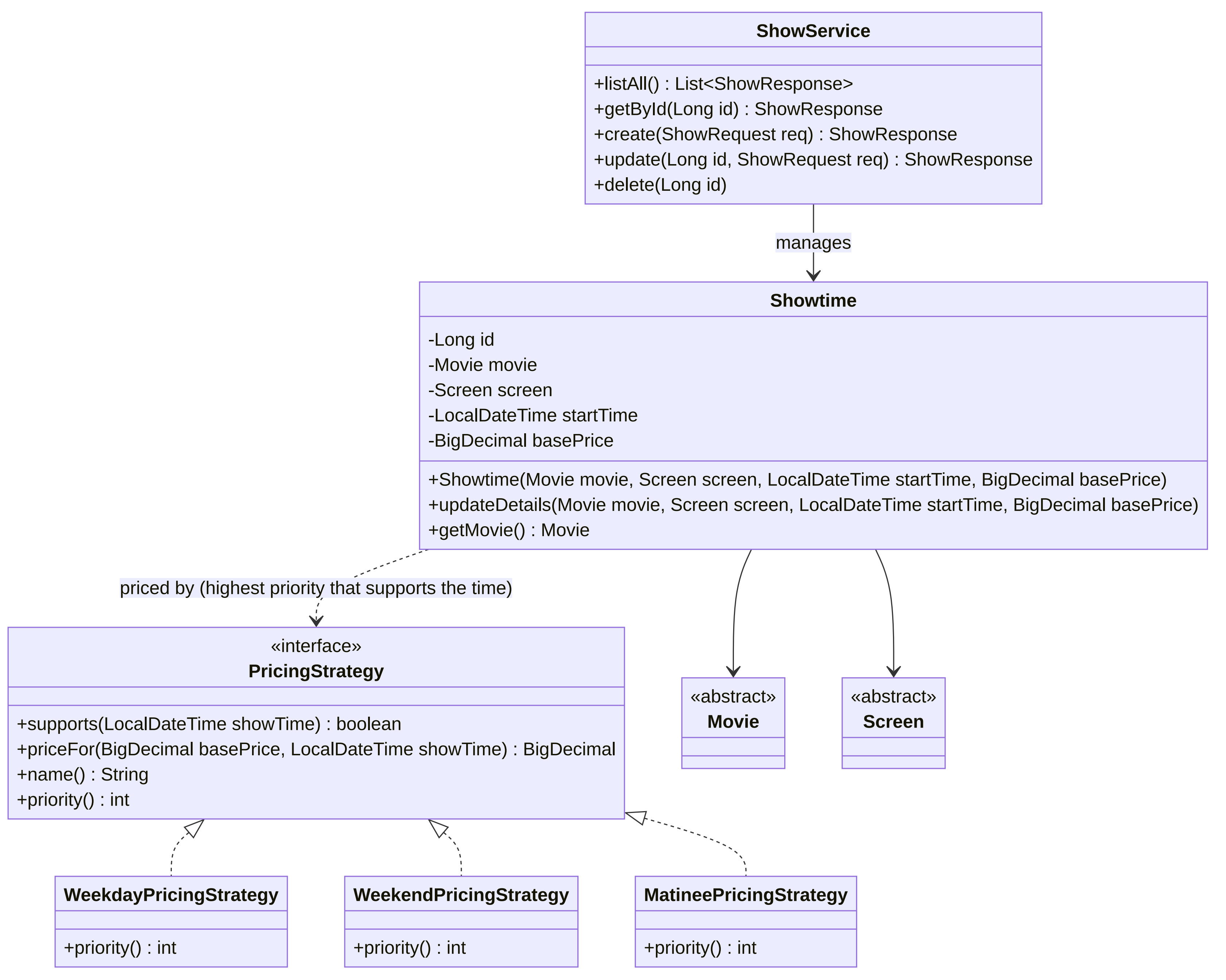
2. Movie catalogue (Member 2)



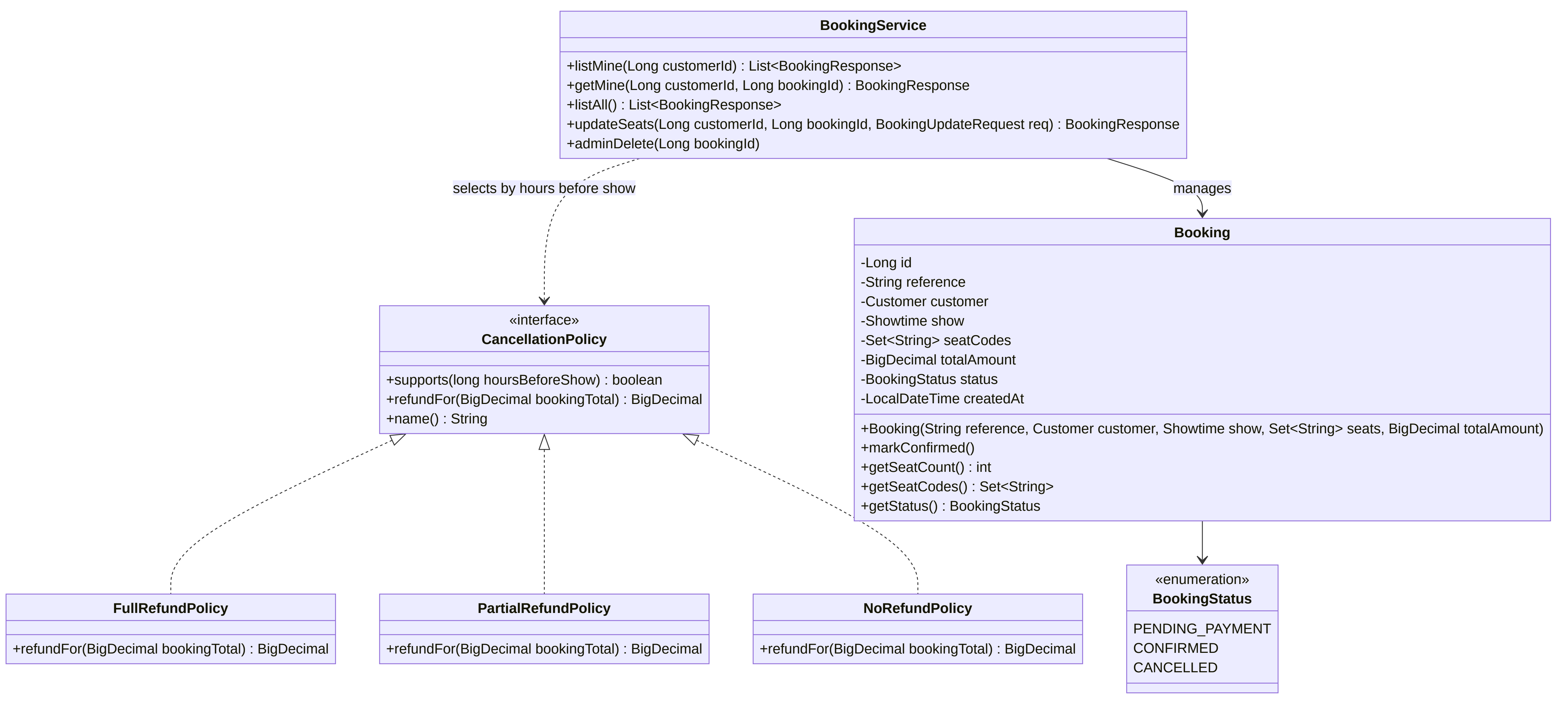
3. Theatres and screens (Member 3)


TheatreService
+list() : List<TheatreResponse>
+get(Long id) : TheatreResponse
+update(Long id, TheatreRequest req) : TheatreResponse
+delete(Long id)
+findEntity(Long id) : Theatre
manages
Theatre
-Long id
-String name
-String city
-String address
-String phone
has
1
*
«abstract»
Screen
-Long id
-String name
-Integer rowCount
-Integer seatsPerRow
-Theatre theatre
+getScreenType() : String
+getPriceMultiplier() : BigDecimal
+getCategory() : String
+setLayout(int rows, int seatsPerRow)
+getTotalSeats() : int
+isValidSeat(String seatCode) : boolean
+getAllSeatCodes() : List<String>
StandardScreen
+getPriceMultiplier() : BigDecimal
PremiumScreen
+getPriceMultiplier() : BigDecimal
VipScreen
+getPriceMultiplier() : BigDecimal
ScreenService
+list(Long theatreId) : List<ScreenResponse>
+get(Long id) : ScreenResponse
+create(ScreenRequest req) : ScreenResponse
+update(Long id, ScreenRequest req) : ScreenResponse
+delete(Long id)
uses
«factory»
ScreenFactory
+create(String screenType) : Screen
creates

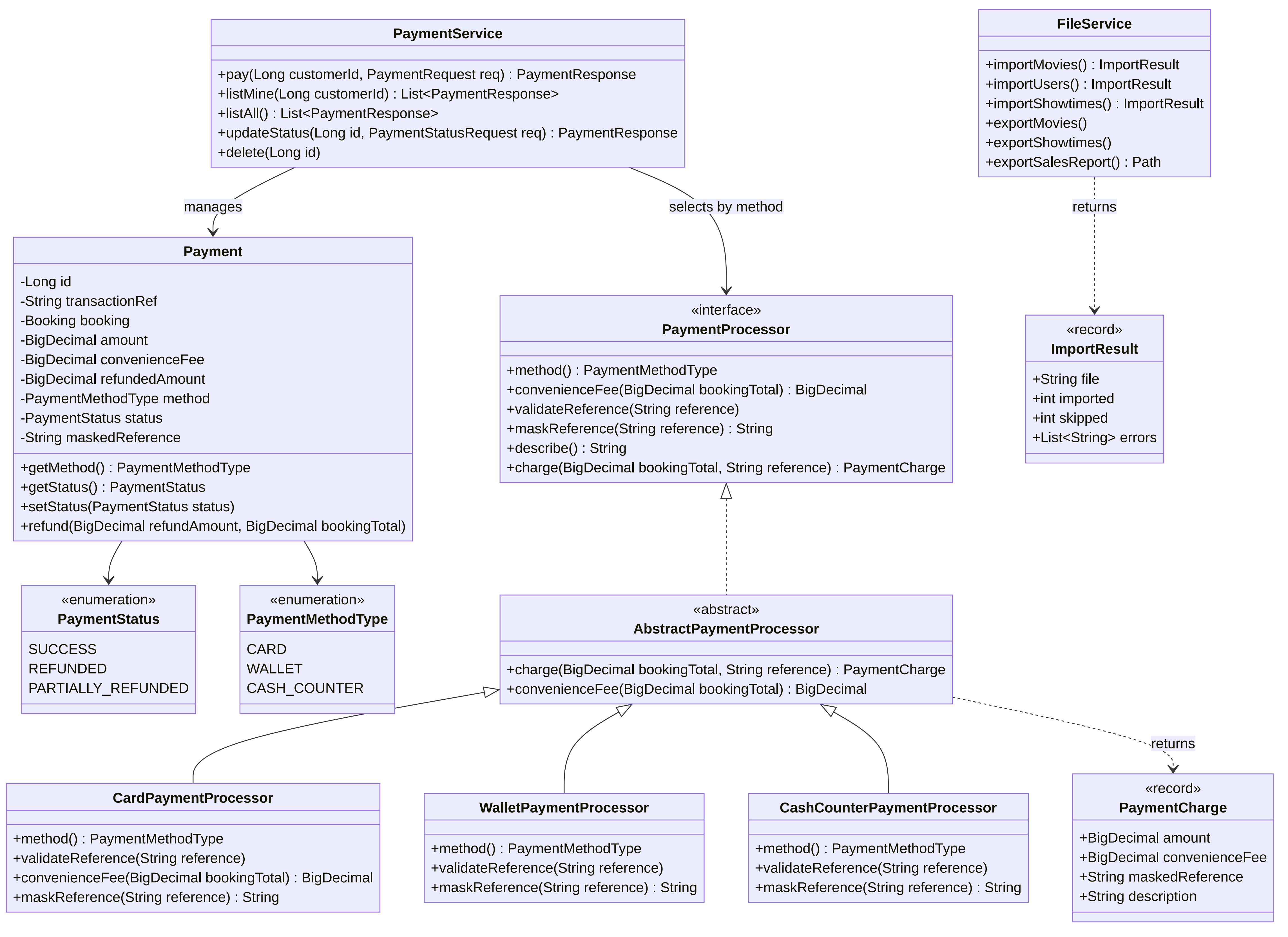
4. Showtime scheduling and pricing (Member 4)


ShowService
+listAll() : List<ShowResponse>
+getById(Long id) : ShowResponse
+create(ShowRequest req) : ShowResponse
+update(Long id, ShowRequest req) : ShowResponse
+delete(Long id)
manages
Showtime
-Long id
-Movie movie
-Screen screen
-LocalDateTime startTime
-BigDecimal basePrice
+Showtime(Movie movie, Screen screen, LocalDateTime startTime, BigDecimal basePrice)
+updateDetails(Movie movie, Screen screen, LocalDateTime startTime, BigDecimal basePrice)
+getMovie() : Movie
priced by (highest priority that supports the time)
«interface»
PricingStrategy
+supports(LocalDateTime showTime) : boolean
+priceFor(BigDecimal basePrice, LocalDateTime showTime) : BigDecimal
+name() : String
+priority() : int
«abstract»
Movie
«abstract»
Screen
WeekdayPricingStrategy
+priority() : int
WeekendPricingStrategy
+priority() : int
MatineePricingStrategy
+priority() : int

5. Booking and seats (Member 5)


BookingService
+listMine(Long customerId) : List<BookingResponse>
+getMine(Long customerId, Long bookingId) : BookingResponse
+listAll() : List<BookingResponse>
+updateSeats(Long customerId, Long bookingId, BookingUpdateRequest req) : BookingResponse
+adminDelete(Long bookingId)
selects by hours before show
manages
«interface»
CancellationPolicy
+supports(long hoursBeforeShow) : boolean
+refundFor(BigDecimal bookingTotal) : BigDecimal
+name() : String
Booking
-Long id
-String reference
-Customer customer
-Showtime show
-Set<String> seatCodes
-BigDecimal totalAmount
-BookingStatus status
-LocalDateTime createdAt
+Booking(String reference, Customer customer, Showtime show, Set<String> seats, BigDecimal totalAmount)
+markConfirmed()
+getSeatCount() : int
+getSeatCodes() : Set<String>
+getStatus() : BookingStatus
FullRefundPolicy
+refundFor(BigDecimal bookingTotal) : BigDecimal
PartialRefundPolicy
+refundFor(BigDecimal bookingTotal) : BigDecimal
NoRefundPolicy
+refundFor(BigDecimal bookingTotal) : BigDecimal
«enumeration»
BookingStatus
PENDING_PAYMENT
CONFIRMED
CANCELLED

6. Payments and file handling (Member 6)


PaymentService
+pay(Long customerId, PaymentRequest req) : PaymentResponse
+listMine(Long customerId) : List<PaymentResponse>
+listAll() : List<PaymentResponse>
+updateStatus(Long id, PaymentStatusRequest req) : PaymentResponse
+delete(Long id)
FileService
+importMovies() : ImportResult
+importUsers() : ImportResult
+importShowtimes() : ImportResult
+exportMovies()
+exportShowtimes()
+exportSalesReport() : Path
manages
selects by method
returns
Payment
-Long id
-String transactionRef
-Booking booking
-BigDecimal amount
-BigDecimal convenienceFee
-BigDecimal refundedAmount
-PaymentMethodType method
-PaymentStatus status
-String maskedReference
+getMethod() : PaymentMethodType
+getStatus() : PaymentStatus
+setStatus(PaymentStatus status)
+refund(BigDecimal refundAmount, BigDecimal bookingTotal)
«interface»
PaymentProcessor
+method() : PaymentMethodType
+convenienceFee(BigDecimal bookingTotal) : BigDecimal
+validateReference(String reference)
+maskReference(String reference) : String
+describe() : String
+charge(BigDecimal bookingTotal, String reference) : PaymentCharge
«record»
ImportResult
+String file
+int imported
+int skipped
+List<String> errors
«enumeration»
PaymentStatus
SUCCESS
REFUNDED
PARTIALLY_REFUNDED
«enumeration»
PaymentMethodType
CARD
WALLET
CASH_COUNTER
«abstract»
AbstractPaymentProcessor
+charge(BigDecimal bookingTotal, String reference) : PaymentCharge
+convenienceFee(BigDecimal bookingTotal) : BigDecimal
returns
CardPaymentProcessor
+method() : PaymentMethodType
+validateReference(String reference)
+convenienceFee(BigDecimal bookingTotal) : BigDecimal
+maskReference(String reference) : String
WalletPaymentProcessor
+method() : PaymentMethodType
+validateReference(String reference)
+maskReference(String reference) : String
CashCounterPaymentProcessor
+method() : PaymentMethodType
+validateReference(String reference)
+maskReference(String reference) : String
«record»
PaymentCharge
+BigDecimal amount
+BigDecimal convenienceFee
+String maskedReference
+String description

7. Cross-module relationships (overview)